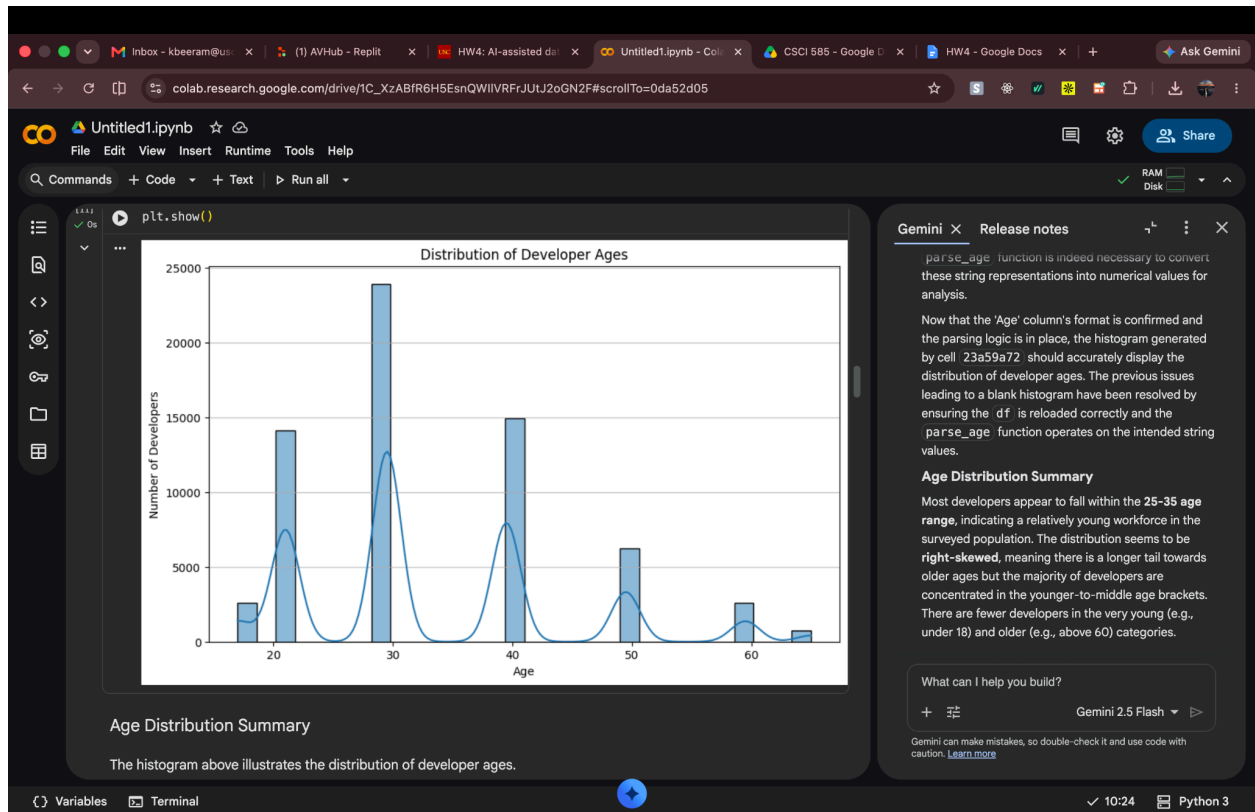
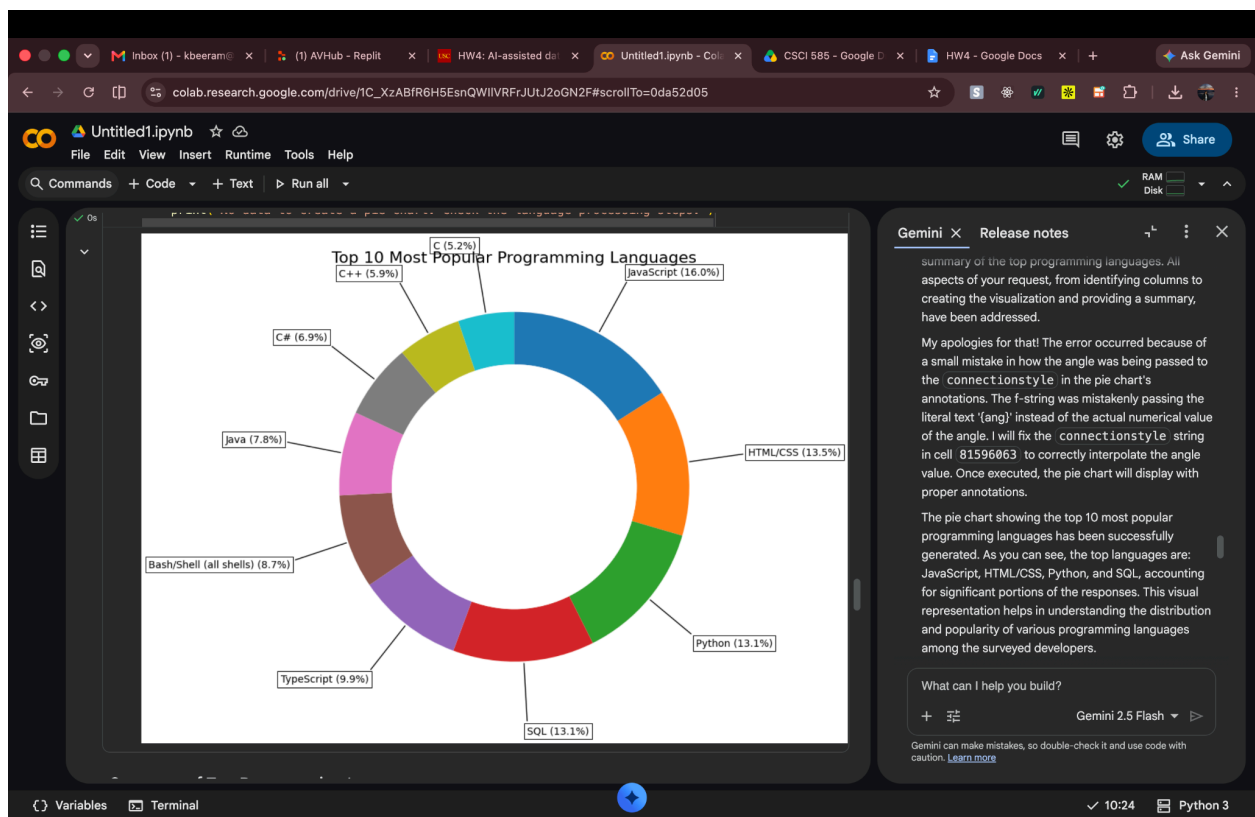


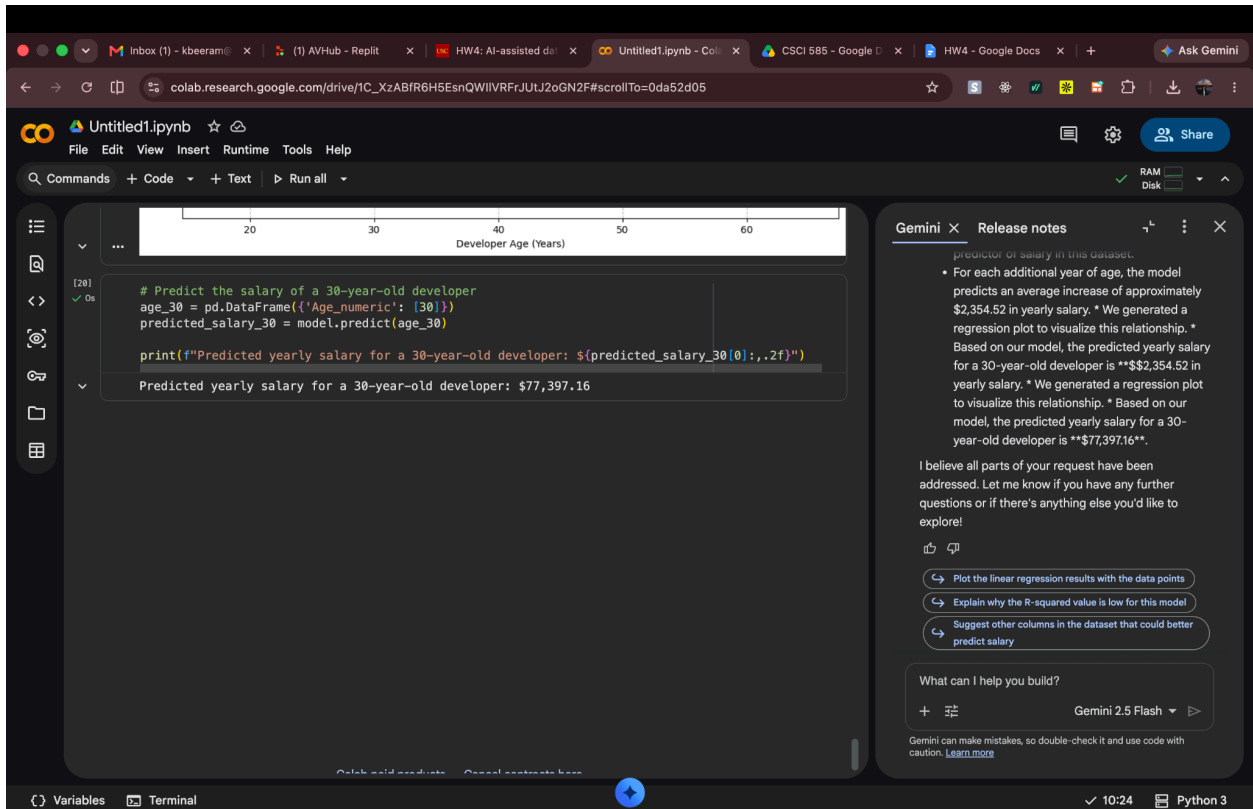
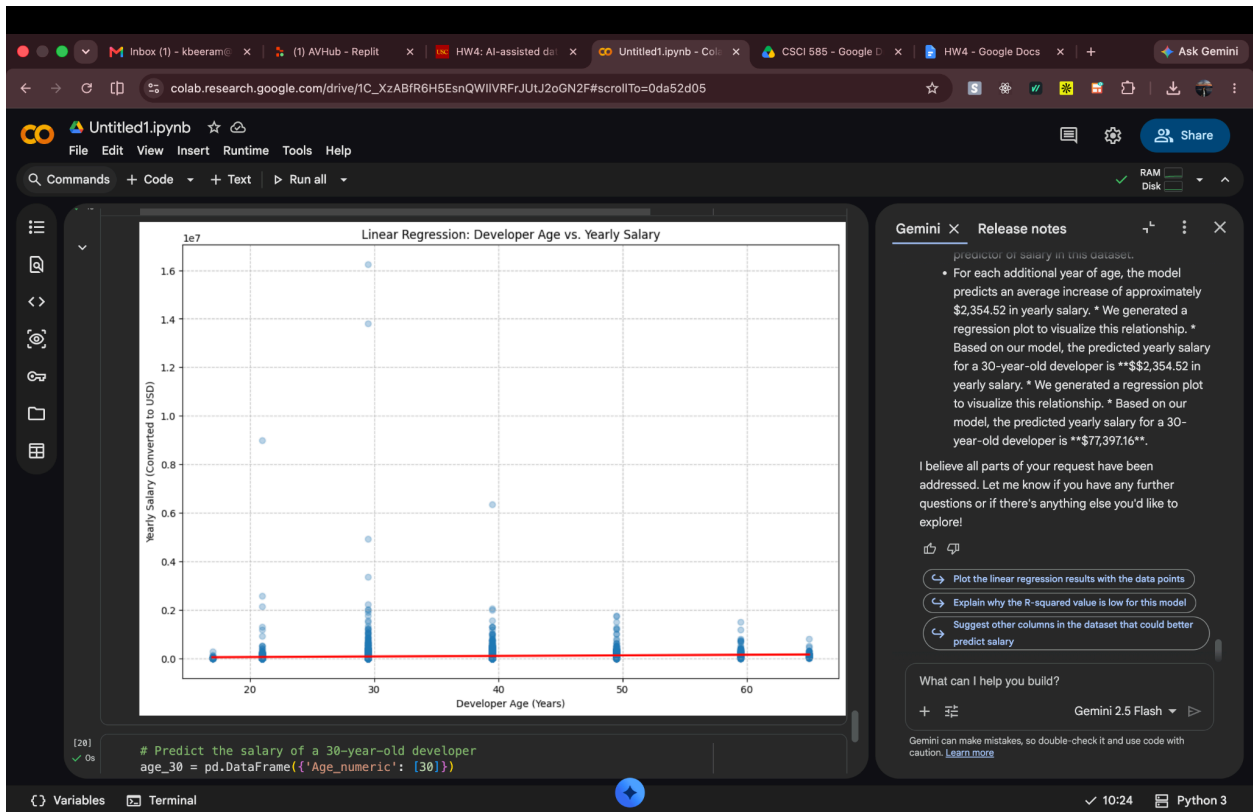
Q1



Q2



Q3



Q4

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Untitled1.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

```
[22] ✓ Os
print(f"Error: Could not decode JSON from '{json_file_path}'. Please check file integrity.")
except Exception as e:
    print(f"An unexpected error occurred: {e}")
```

Values under the 'pins' key:

Pin 1:

```
name: D0
port: D
num: 0
sortbyname: D00
functions: {'XL1': 0, '3.3': 0, 'NO_BLOCKLY': 0}
csv: {}
simplefunctions: {}
```

Pin 2:

```
name: D1
port: D
num: 1
sortbyname: D01
functions: {'XL2': 0, '3.3': 0, 'NO_BLOCKLY': 0}
csv: {}
simplefunctions: {}
```

Pin 3:

```
name: D2
port: D
num: 2
sortbyname: D02
functions: {'ADC1_IN0': 0, '3.3': 0, 'NO_BLOCKLY': 0, 'PRESSURE': 'pin_scl'}
csv: {}
simplefunctions: {'ADC': ['ADC1_IN0']}
```

Pin 4:

```
name: D3
num: 3
sortbyname: D03
functions: {'ADC1_IN1': 0, '3.3': 0, 'NO_BLOCKLY': 0, 'BAT': 'pin_voltage'}
```

Gemini Release notes

Load the uploaded BANGLEJS2.json file, list all values under the pins key in a readable format.

Okay, I will load the BANGLEJS2.json file and display all values under the 'pins' key.

You're right! It looks like the pins key contains a list of pin configurations, not a dictionary. I'll correct the code to iterate through the list and display each pin's details in a readable format.

The BANGLEJS2.json file has been loaded, and all values under the 'pins' key have been extracted and displayed in a readable format. You can see the details for each pin, including its name, port, number, sorting name, functions, csv data, and simple functions.

Filter the 'pins' data to show only those with 'HEARTRATE' functionality

Count how many pins serve the 'LCD' device

Create a summary table of all pin functions available in the JSON file

What can I help you build?

+ Gemini 2.5 Flash

Gemini can make mistakes, so double-check it and use code with caution. [Learn more](#)

Q5

colab.research.google.com/drive/1C_XzABfR6H5EsnQWIIvRfrJULJ2oGN2F#scrollTo=594b42b8

Untitled1.ipynb

File Edit View Insert Runtime Tools Help

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```
[23] ✓ Os
print("The 'devices' key was not found in the JSON file.")
```

Values under the 'devices' key:

Device: BTN1

```
pin: D17
pinstate: IN_PULLDOWN
```

Device: LED1

```
pin: D8
novariable: True
```

Device: LCD

```
width: 176
height: 176
bpp: 3
controller: LPM013M126
pin_cs: D5
pin_extcomin: D6
pin_disp: D7
pin_sck: D26
pin_mosi: D27
pin_bl: D8
```

Device: TOUCH

```
device: CST8165
addr: 21
pin_sda: D33
pin_scl: D34
pin_rst: D35
pin_irq: D36
```

Device: VIBRATE

```
pin: D19
```

Device: GPS

```
device: Casic URANUS
pin_en: D29
pin_rx: D30
```

Gemini Release notes

values under the 'devices' key, and describe the devices has already been added to your notebook. Please run the new cells to see the output.

the code is not there

My apologies, it seems the code for extracting and describing the 'devices' key was not added to your notebook as intended. I will add it now, along with a brief description of the devices.

The code has successfully extracted and displayed the values under the 'devices' key from the BANGLEJS2.json file. You can now see the detailed specifications for various components such as BTN1, LED1, LCD, TOUCH, VIBRATE, GPS, BAT, HEARTRATE, ACCEL, MAG, PRESSURE, and SPIFLASH, including their pins, addresses, and other relevant properties.

Filter for devices that require 'sda' and 'scl' pins

What can I help you build?

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